Seasonal Moving Linear Scaling (SMLS) Model
Documentation

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1 Model Overview

The **Seasonal Moving Linear Scaling (SMLS)** model is a simple, data-driven forecasting method for photovoltaic (PV) power generation. The model uses historical PV production and irradiance measurements from previous days—aligned at the same time of day—to estimate a linear relationship between solar radiation and PV output. This relationship is then applied to future irradiance forecasts to obtain the PV power prediction.

The approach exploits the strong daily periodicity of solar power systems and the near-proportionality between irradiance and PV output under stable operating conditions. It requires no information about installation-specific parameters such as panel orientation, tilt, or capacity, and serves as a robust baseline method.

2 Method Description

2.1 Forecast Target

Let t_0 denote the time to be forecasted (e.g., tomorrow at 09:00). The dataset has a time resolution of 15 minutes, so each day corresponds to 96 time steps.

2.2 Seasonal Lags

Historical timestamps t_l are selected such that they correspond to the same time of day as t_0 but from previous days:

$$t_l = t_0 - 96 \cdot i, \quad i = 1, 2, \ldots, 12.$$

For each lag, we collect:

$$PV[t_l] \quad \text{and} \quad S[t_l],$$

representing the measured PV power and solar irradiance, respectively.

These form the set of aligned historical samples:

$$\mathcal{D} = \{(S[t_l],\, PV[t_l]) : l \in \text{lags}\}\,.$$

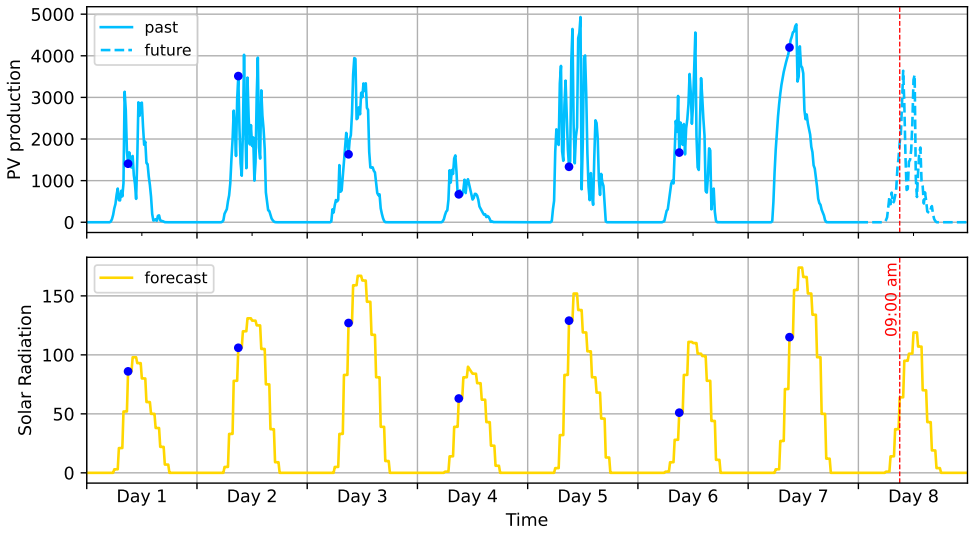


Figure 1: Example timeseries for PV production mesurements (top) and solar radiation forecast (bottom). Restricted to 8 days in the past for cleaner visualization. We assume we are creating forecasts at the beginning of day 8. As an example we high-light 09:00 am as timestamp which we want to forecast.

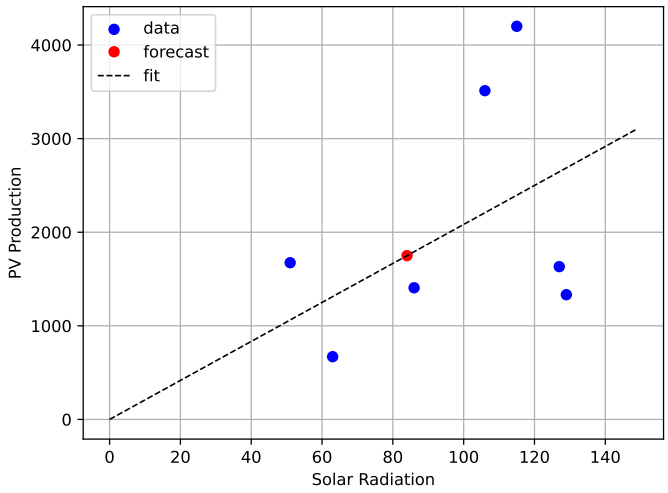


Figure 2: Representation of the linear interpolation for one timestamp (09:00 am).

2.3 Linear Scaling Coefficient

The model assumes a linear relationship without an intercept:

$$PV \approx m \cdot S.$$

The scaling factor m is estimated by:

$$m = \frac{\text{mean}(S[t_l] \cdot PV[t_l])}{\text{mean}(S[t_l]^2)}.$$

This corresponds to a least-squares fit of a one-parameter linear model (no bias term).

2.4 Forecasting

Given the forecasted irradiance $S[t_0]$ from a weather model, the PV forecast is:

$$\widehat{PV}[t_0] = m \cdot S[t_0].$$

3 Why the Model Works

3.1 Daily Seasonality

PV production at a given time of day tends to be repeatable across days, reflecting consistent solar geometry and typical shading conditions. By using samples from the same time of day on previous days, the model implicitly captures effects such as:

- sun angle,
- orientation and tilt of the installation,
- shading patterns,
- system-specific performance.

3.2 Linear Relationship Between Irradiance and PV Output

For a fixed geometry, PV power is approximately proportional to the incident irradiance. The coefficient m implicitly encodes:

- system size (kWp),
- orientation and tilt,
- inverter clipping behavior (partially),
- local shading and system losses.

Thus, the model learns the site-specific mapping from irradiance to PV output without requiring explicit physical metadata.

3.3 Noise Robustness

Averaging across many previous days stabilizes the estimate of m, reducing the impact of:

- measurement noise,
- short-term irradiance fluctuations,
- temporary anomalies in PV production.

3.4 Simplicity and Efficiency

The SMLS model is computationally lightweight, easy to implement, and automatically adapts to slow performance drifts in the PV system. It provides a reliable baseline forecast and can be used alongside more sophisticated models.

4 Summary

The Seasonal Moving Linear Scaling (SMLS) model forecasts PV power using an intuitive and interpretable approach:

1. Collect irradiance and PV values from the same time of day over previous days.
2. Fit a no-intercept linear model to estimate the irradiance–power scaling factor.
3. Apply this scaling factor to forecasted irradiance to obtain the PV power prediction.

Despite its simplicity, the model leverages strong periodicity in solar production and the fundamental physical link between irradiance and PV output, making it an effective baseline forecasting method.